

AI-Powered CRM Automation & Intelligent Lead Qualification System

Project Overview

This project is a complete production-style AI-powered CRM Automation System built using:

- n8n
- Groq AI
- Gemini
- Google Sheets
- Gmail
- ClickUp
- Webhooks
- HTML/CSS/JavaScript

The system automates the complete lead lifecycle from lead capture to AI qualification, CRM management, follow-up tracking, lead routing, and internal team notifications.

This project combines:

- CRM automation
- AI lead analysis
- AI qualification
- Lead scoring
- Lead routing
- Activity tracking
- Follow-up automation
- Smart notifications
- Production-level workflow architecture

The goal of this system is to simulate a real-world business CRM automation environment using scalable and modular workflow design.

Main Business Objectives

This automation system was designed to solve real CRM problems such as:

- Manual lead management
- Slow sales response
- Duplicate lead creation
- Poor follow-up tracking
- Missing high-value leads
- Lack of lead prioritization
- No centralized activity tracking

The system automates all these processes using workflows and AI.

Technologies Used

Technology	Purpose
n8n	Workflow automation
Groq AI & Gemini	AI processing & qualification
Google Sheets	CRM database
Gmail	Customer & internal emails
ClickUp	Internal notifications
HTML/CSS/JavaScript	Frontend forms
Webhooks	Frontend-backend communication

Complete System Architecture

The complete automation system contains 3 workflows.

Workflow	Purpose
Workflow 1	AI Lead Capture & Qualification
Workflow 2	CRM Management & Activity Tracking
Workflow 3	Automated Follow-Up Reminder System

Each workflow handles a separate business operation.

This modular architecture follows real production workflow design.

Google Sheets Database Structure

The system uses 2 sheets.

Sheet 1 — CRM_Leads

This sheet stores all lead records.

Column Name
Lead_ID
Name
Email
Phone
Company
Service
Budget
Urgency
Source
Message
AI_Summary
Lead_Category
Lead_Intent
Lead_Quality
AI_Lead_Score
AI_Confidence_Score
Status
Last_Activity
Created_At
Updated_At
Next_Follow_Up

Sheet 2 — Activity_Logs

This sheet stores CRM activity history.

Column Name
Activity_ID
Lead_ID
Activity_Type
Status_After_Update
Notes
Follow_Up_Date
Activity_Time

Pakistan Standard Time Configuration

The entire system uses Pakistan Standard Time.

Date & Time Format

```
{{
    DateTime.now()
    .setZone('Asia/Karachi')
    .toFormat('dd LLL yyyy, hh:mm:ss a')
}}
```

Example Output

13 May 2026, 10:35:26 AM

This format is used for:

- Created_At
- Updated_At
- Activity_Time

Lead ID Generation Logic

Lead IDs are generated only when a completely new lead is created.

Lead ID Format

```
LD-{{ Math.random().toString(36).substring(2,6).toUpperCase() }}
```

Example Output

LD-A7F2

Why Lead IDs Are Generated Only for New Leads

Originally, Lead IDs were generated before duplicate checking.

That created several problems:

- Duplicate leads generated fake new IDs
- Slack notifications showed incorrect IDs

- Email notifications displayed mismatched IDs
- CRM records became inconsistent

To solve this professionally:

- Lead IDs are generated only inside the “Store New Lead in Database” node
- Existing leads always keep their original IDs

This follows real production CRM architecture.

Activity ID Generation Logic

Activity IDs are generated whenever an admin updates a lead.

Activity ID Format

```
ACT-{{ Math.random().toString(36).substring(2,6).toUpperCase() }}
```

Example Output

```
ACT- -K2P8
```

WORKFLOW 1

AI Lead Capture & Qualification Workflow

Workflow Purpose

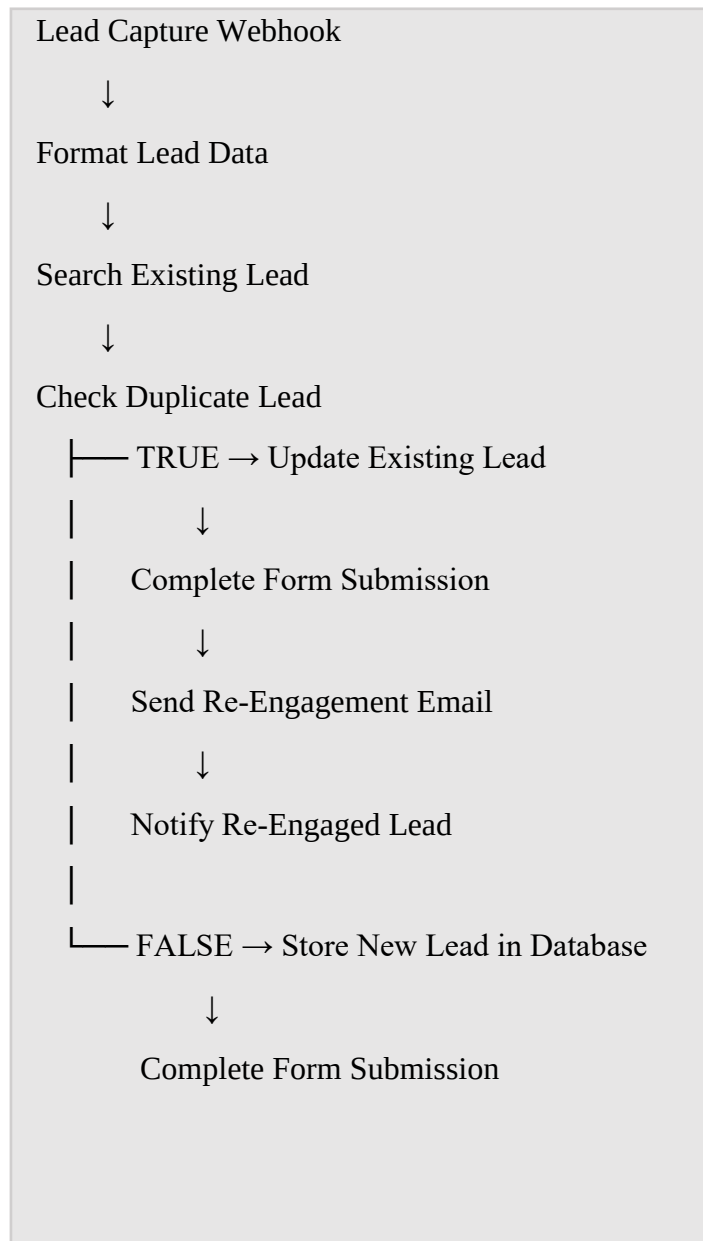
This workflow:

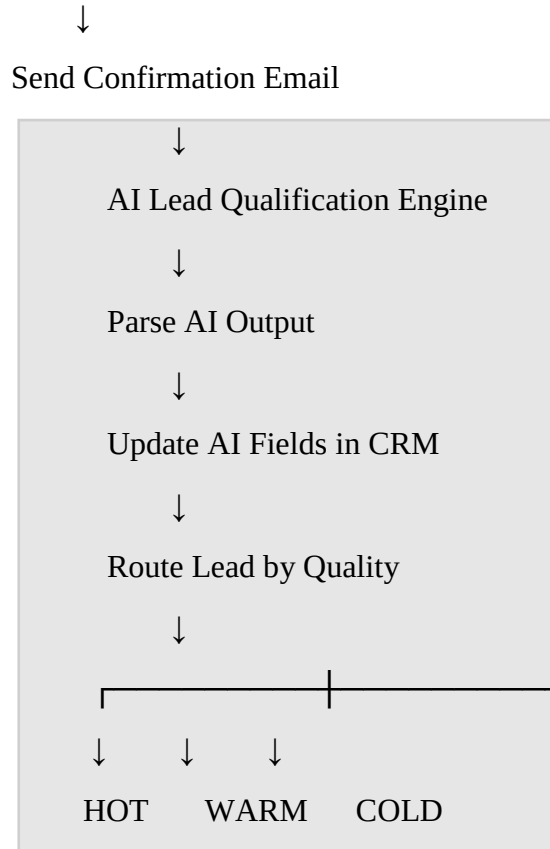
- captures leads from frontend form
- checks duplicate leads
- updates existing leads
- stores new leads
- processes AI qualification
- generates AI summary
- categorizes leads
- detects buying intent

- calculates lead quality
- calculates lead score
- generates confidence score
- routes leads automatically
- Creates ClickUp tasks.
- sends internal emails
- sends customer emails

This workflow acts as the main AI-powered lead intake system.

Workflow 1 Structure





Workflow 1 Node Documentation

1. Lead Capture Webhook

Purpose

Receives lead form submissions from frontend form.

Function

- accepts POST request
- receives form data
- triggers workflow execution

2. Format Lead Data

Purpose

Formats incoming lead information.

Function

- cleans form values
- maps fields correctly
- creates timestamps
- prepares CRM structure

3. Search Existing Lead

Purpose

Checks whether lead already exists.

Duplicate Matching Logic

The workflow checks duplicates using:

- Email
- Service

Why Email + Service?

Using only email is not realistic.

Because:

- same customer may request multiple services
- different services should create separate leads

Professional CRM Logic

Email	Service	Result
ali@gmail.com	AI Automation	Existing Lead
ali@gmail.com	Web Development	New Lead

This follows realistic CRM behavior.

4. Check Duplicate Lead

Purpose

Determines whether:

- lead already exists
OR
 - lead is completely new
-

TRUE Path — Existing Lead

5. Update Existing Lead

Purpose

Updates existing CRM record.

Function

- updates latest message
- updates timestamps
- updates activity tracking
- keeps original Lead_ID

No new Lead ID is generated.

6. Complete Form Submission

Purpose

Returns success response to frontend form.

Success Response

```
{  
  "success": true,  
}
```

7. Send Re-Engagement Email

Purpose

Sends email when duplicate lead becomes active again.

This improves customer communication.

8. Notify Re-Engaged Lead

Purpose

Creates ClickUp task for internal team.

This helps the sales team prioritize active leads.

FALSE Path — New Lead

9. Store New Lead in Database

Purpose

Creates completely new CRM lead.

Functions

- generates Lead_ID
- stores lead
- sets timestamps
- sets default status
- generates next follow-up date

Automatic Follow-Up Date Logic

The system automatically creates the next follow-up date.

Expression

```
{{  
  DateTime.now()  
    .setZone('Asia/Karachi')  
    .plus({ days: 1 })  
    .toFormat('dd LLL yyyy')  
}}
```

Example Output

14 May 2026

10. Send Confirmation Email

Purpose

Sends professional confirmation email to customer.

This improves trust and communication.

11. AI Lead Qualification Engine

Node Type

Basic LLM Chain

AI Model Used

Groq or Gemini

AI Processing Features

The AI automatically performs:

- lead summarization
- lead categorization
- lead intent analysis
- lead quality classification
- lead scoring
- confidence analysis

AI Definitions

Feature	Meaning
AI Summary	Short summary of lead
Lead Category	Type of requested service
Lead Intent	Buying intent
Lead Quality	Hot / Warm / Cold
AI Lead Score	Lead value score
Confidence Score	AI confidence level

Lead Intent Meaning

Intent	Meaning
High	Strong buying interest
Medium	Moderate interest

Low	Exploring only
-----	----------------

Lead Quality Meaning

Quality	Meaning
Hot	High-value lead
Warm	Moderate-value lead
Cold	Low-priority lead

Difference Between Lead Intent & Lead Quality

Lead Intent

Shows customer buying interest.

Lead Quality

Shows overall business value of the lead.

Both are different and both are important in real businesses.

AI Temperature

0.3

Lower temperature gives more stable business responses.

Max Tokens

500

AI System Prompt

You are an AI CRM Qualification Assistant.
Analyze the lead information carefully.
Your tasks:

1. Create a short professional summary.
2. Detect lead category.
3. Detect buying intent.
4. Classify lead quality as Hot, Warm, or Cold.
5. Generate lead score from 1-100.
6. Generate confidence score from 1-100.

Rules:

HOT Lead:

- High budget
- Urgent timeline
- Clear business need
- Serious inquiry

WARM Lead:

- Moderate budget
- Some interest
- Needs nurturing

COLD Lead:

- Low urgency
- Low budget
- Exploring only

Return ONLY valid JSON.

Format:

```
{  
  "summary": "",  
  "category": "",  
  "intent": "",  
  "lead_quality": "",  
  "lead_score": "",  
  "confidence_score": ""  
}
```

AI User Prompt

Lead Information:

Name: {{ \$json.Name }}

Service Interested In: {{ \$json.Service }}

Budget: {{ \$json.Budget }}

Urgency: {{ \$json.Urgency }}

Message:

{{ \$json.Message }}

Analyze this lead.

8. AI Response Parsing

Problem

Groq and Gemini sometimes return:

- markdown
- wrapped JSON
- inconsistent formatting

Solution

A custom JavaScript Code node parser was implemented.

This provides:

- better reliability

- easier debugging
- production-safe architecture

Code Node — Parse AI Response

Code Used

```
let raw = "";

if ($json.text) {
  raw = $json.text;
}
else if ($json.response?.generations?.[0]?.[0]?.text) {
  raw = $json.response.generations[0][0].text;
}

raw = raw
  .replace(/``json/g, "")
  .replace(/```/g, "")
  .trim();

const parsed = JSON.parse(raw);

return {
  json: parsed
};
```

13. Format AI Output

Purpose

Extracts AI fields from JSON response.

Extracted Fields

Field
AI_Summary
Lead_Category
Lead_Intent
Lead_Quality
AI_Lead_Score
AI_Confidence_Score

14. Route Lead by Quality

Node Type

Switch Node

Routing Logic

Lead Quality	Route
Hot	Hot Lead Notifications
Warm	Warm Lead Notifications
Cold	Cold Lead Notifications

15. Lead Quality Routing

Hot Lead Path

Actions

- Create ClickUp task
- Send Gmail alert
- Apply Gmail label

Gmail Label

Hot Leads

Warm Lead Path

Actions

- Create ClickUp task
- Send Gmail alert
- Apply Gmail label

Gmail Label

Warm Leads

Cold Lead Path

Actions

- Send Gmail alert only

- Apply Gmail label

Gmail Label

Cold Leads

16. ClickUp Integration

Purpose

Used for internal team collaboration and sales task management.

ClickUp Lists Used

List	Purpose
Hot Leads	High-priority leads
Warm Leads	Medium-priority leads
Follow-Ups & Updates	CRM updates and reminders

17. ClickUp Task Configuration

Hot Lead Task

Priority

Urgent

Status

New Lead

Task Content Includes

- Lead details
- AI summary
- lead score
- lead intent
- urgency
- budget

Warm Lead Task

Priority

Normal

Status

Qualified

Follow-Up Reminder Task**Priority**

High

Status

Follow-Up Required

18. Gmail Automation**Emails Sent**

Email Type	Purpose
Customer Confirmation	Confirms lead submission
Hot Lead Alert	Internal notification
Warm Lead Alert	Internal notification
Cold Lead Alert	Internal notification
Follow-Up Reminder	Follow-up notification

14. Gmail Label Automation

The workflow automatically applies Gmail labels:

- Hot Leads
- Warm Leads
- Cold Leads
- Lead Follow-Ups

using Gmail Add Label nodes.

WORKFLOW 2

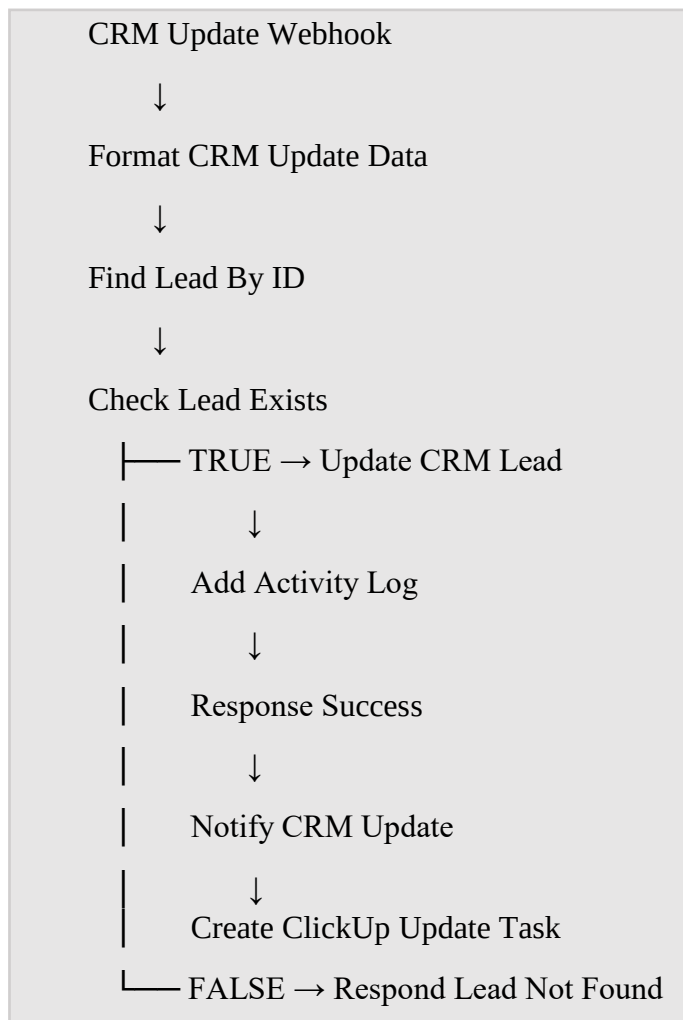
CRM Management & Activity Tracking Workflow

Purpose

This workflow allows internal admins to:

- update CRM leads
- schedule follow-ups
- update statuses
- add notes
- track activities

Workflow 2 Structure



Workflow 2 Features

- secure lead updates
- activity tracking
- follow-up management
- CRM history logging
- backend validation
- professional error handling

Success Response

```
{  
  "success": true,  
}
```

Error Response

```
{  
  "success": false,  
  "message": "Lead ID not found. Please enter a valid Lead ID."  
}
```

Why This Is Important

Without backend responses:

- frontend cannot show toast messages
- users see generic errors
- form UX becomes poor

This creates professional frontend-backend communication.

WORKFLOW 3

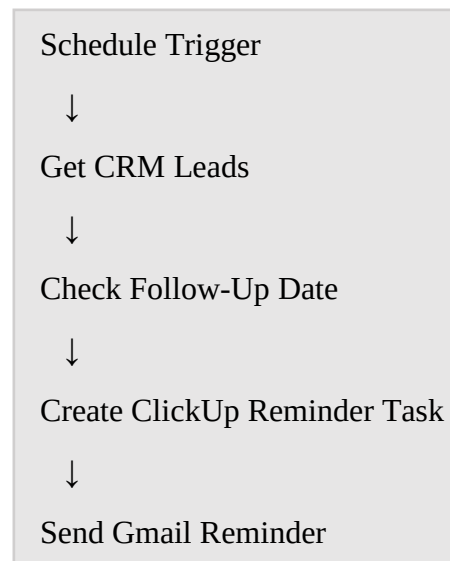
Automated Follow-Up Reminder Workflow

Purpose

This workflow automatically:

- checks overdue follow-ups
- detects inactive leads
- notifies internal team
- prevents missed opportunities

Workflow 3 Structure



Daily Trigger

Schedule

Every Day — 9:00 AM Pakistan Time

Get CRM Leads

Important Configuration

Return All = TRUE

This ensures all leads are processed.

Overdue Detection Logic

The workflow checks:

- lead is not closed
- follow-up date is overdue

Best Production-Level Date Condition

```
{{  
  DateTime.fromFormat(  
    $json["Next_Follow_Up"],  
    "dd LLL yyyy"  
  ).toMillis() < DateTime.now().setZone('Asia/Karachi').toMillis()  
}}
```

Why This Condition Is Best

Because it:

- supports custom date format
- avoids month/day confusion
- works with Pakistan timezone
- follows production logic

Example Overdue Notification

Overdue Follow-Up Alert

Lead ID: LD-A7F2

Client: Ali Khan

Service: AI Automation

Current Status: Proposal Sent

Follow-Up Date: 12 May 2026

Frontend Features

The frontend forms include:

- responsive UI
- validation
- toast notifications
- loading states
- dynamic fields
- professional design
- “Other” option handling

Real Production-Level Form Features

Service Options

- AI Automation
- CRM Automation
- Chatbot Development
- Workflow Automation
- Web Development
- API Integration
- Other

Budget Options

- Less than \$500
- \$500-\$2000
- \$2000-\$5000
- \$5000+
- Other

Urgency Options

- Immediately
- Within 1 Week
- Within 1 Month
- 2-3 Months
- Flexible
- Just Exploring
- Other

Dynamic “Other” Option Logic

When user selects:

Other

A custom text field appears automatically.

The custom value replaces the dropdown value before submission.

This creates cleaner CRM records without additional columns.

Error Handling Features

The system handles:

- invalid Lead IDs
- duplicate requests
- webhook failures
- API failures
- AI formatting issues
- frontend validation errors
- network failures

Production-Level Features

AI Features

- AI qualification
- AI routing
- AI lead scoring
- AI confidence analysis
- AI lead categorization

CRM Features

- duplicate detection
- activity tracking
- follow-up scheduling
- centralized CRM management

Automation Features

- ClickUp task automation
- automated lead routing
- automated qualification
- automated reminders
- automated notifications

Repository Structure

```
ai-powered-crm-automation-system/  
|  
├── workflows/  
|   ├── 01-ai-lead-capture-qualification.json  
|   ├── 02-crm-management-activity-tracking.json  
|   └── 03-followup-reminder-system.json  
|
```

```
├── forms/
│   ├── lead-capture-form.html
│   └── admin-crm-update-form.html
│
├── assets/
│   ├── workflow-1-ai-lead-capture.png
│   ├── workflow-2-crm-management.png
│   ├── workflow-3-followup-reminder.png
│   ├── clickup-tasks.png
│   ├── gmail-messages.png
│   ├── crm-google-sheet.png
│   ├── lead-from.png
│   └── admin-form.png
│
├── documentation/
│   └── project-documentation.md
│
├── README.md
│
└── .gitignore
```

Final Project Outcome

The final system successfully demonstrates:

- AI-powered CRM automation
- intelligent lead qualification
- AI lead scoring
- AI lead routing

- duplicate lead management
- automated follow-up management
- CRM activity tracking
- ClickUp task automation
- Gmail automation
- production-level workflow architecture

Final Conclusion

This project simulates a real-world AI CRM automation environment using:

- workflow automation
- AI processing
- CRM management
- internal collaboration systems

The system helps businesses:

- reduce manual work
- improve lead qualification
- prioritize high-value leads
- improve response speed
- automate follow-ups
- centralize CRM operations

The workflows were designed using scalable and modular production-oriented architecture while remaining easy to maintain and extend.

Important Note

The exported n8n workflow JSON files do NOT contain credentials.

Users must add their own:

- Gmail credentials

- Google Sheets credentials
- ClickUp credentials
- Groq API keys
- Gemini API keys

after importing workflows into their own n8n instance.